

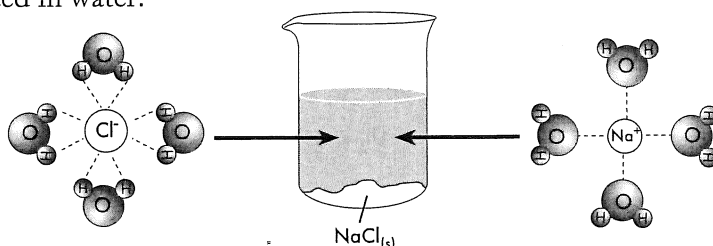
Acids and Bases in Aqueous Solution

5.1 ELECTROLYTES



The size, shape and polarity of water molecules make it possible for them to pack very closely around solute particles (ions or molecules). This makes water a very good solvent.

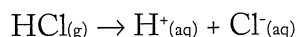
e.g. 1. $\text{NaCl}_{(s)}$ placed in water:



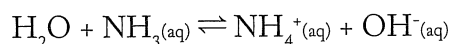
- Negative region of H_2O is attracted to positive Na^+ ion.
- H_2O molecules 'pull' Na^+ ion from the Cl^- ion and disperse the Na^+ ions throughout the solution.
- Positive region of H_2O attracted to Cl^- ions and the same process with Na^+ and Cl^- ions causes $\text{NaCl}_{(s)}$ to dissolve.

Substances that dissolve in water and form ions are called ELECTROLYTES.

STRONG ELECTROLYTES – Are good conductors of electricity in aqueous solution because water causes them to dissociate (ionic substances) or completely ionise (covalent molecular substances). Ionic substances are all strong electrolytes even if only slightly soluble since any amount dissolved completely breaks up into ions. Strong acids are also strong electrolytes as they completely ionise in solution.

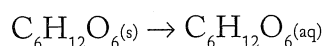


WEAK ELECTROLYTES – Are poor conductors of electricity in aqueous solution. Only a small percentage of molecules ionise when dissolved in water and so only a small concentration of ions is produced to conduct charge. Weak acids, and weak covalent molecular bases (such as NH_3), tend to be weak electrolytes.

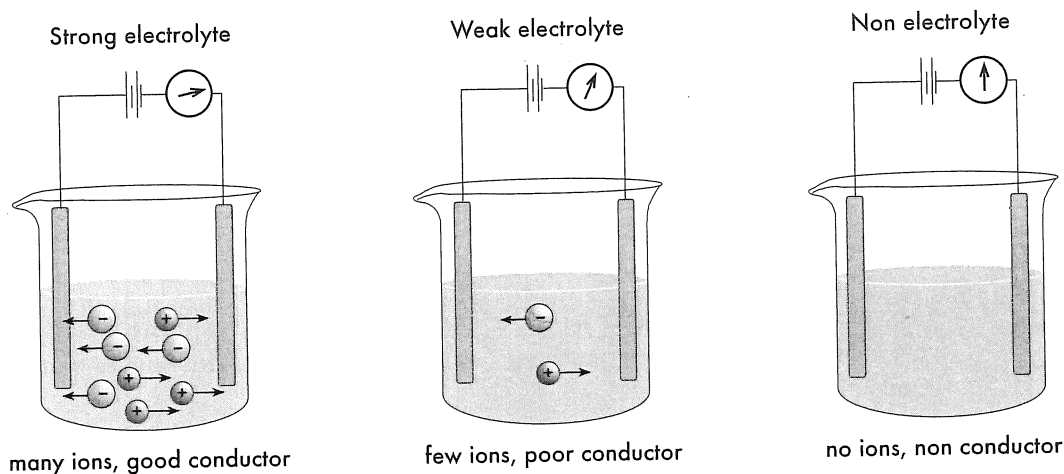


The double arrow ($\rightleftharpoons$) is used to indicate that only a small percentage of molecules ionise.

NON ELECTROLYTES – Are non-conductors of electricity because no ions are produced when they dissolve in water.

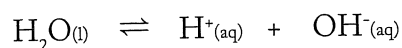


Glucose molecules dissolve but no ions are formed.



Self-Ionisation of Water

Water, no matter how pure, is always a very weak conductor of electricity. This is due to the self ionisation of water.



A 1.00 litre sample of pure water at 25°C contains 1.00×10^{-7} mol of H^+ ions and 1.00×10^{-7} mol of OH^- ions.

i.e. at 25°C, $[\text{H}^+] = [\text{OH}^-] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$ and because $[\text{H}^+] = [\text{OH}^-]$ the water is said to be neutral.

The equilibrium constant for this self ionisation is given the symbol K_w .

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

The only factor that effects K_w is the temperature. As temperature rises, so does K_w but if it is pure water, no matter what the value of K_w , $[\text{OH}^-] = [\text{H}^+]$.

ACIDIC SOLUTIONS: $[\text{H}^+] > [\text{OH}^-]$

or $[\text{H}^+] > 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] < 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

BASIC SOLUTIONS: $[\text{H}^+] < [\text{OH}^-]$

or $[\text{H}^+] < 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] > 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

NEUTRAL SOLUTIONS: $[\text{H}^+] = [\text{OH}^-]$

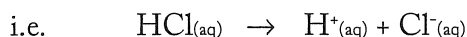
or $[\text{H}^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

and $[\text{OH}^-] = 1.00 \times 10^{-7} \text{ mol L}^{-1} \text{ at } 25^\circ\text{C}$

Worked Examples

- 5.1 Calculate the $[\text{OH}^-]$ of a $2.00 \times 10^{-2} \text{ mol L}^{-1}$ solution of HCl

NB: HCl is a strong acid and will completely ionise



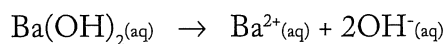
$$\text{therefore} \quad [\text{H}^+] = [\text{HCl}] = 2.00 \times 10^{-2} \text{ mol L}^{-1}$$

$$K_w = [\text{H}^+] [\text{OH}^-]$$

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.00 \times 10^{-14}}{2.00 \times 10^{-2}}$$

$$[\text{OH}^-] = 5.00 \times 10^{-13} \text{ mol L}^{-1}$$

- 5.2 Calculate the $[\text{H}^+]$ of a 0.350 mol L^{-1} solution of $\text{Ba}(\text{OH})_2$.



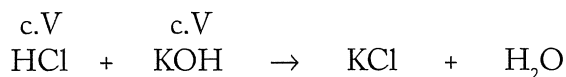
$$[\text{OH}^-] = 2 \times [\text{Ba}(\text{OH})_2] = 0.700 \text{ mol L}^{-1}$$

$$K_w = [\text{H}^+] [\text{OH}^-]$$

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{0.700}$$

$$[\text{H}^+] = 1.43 \times 10^{-14} \text{ mol L}^{-1}$$

- 5.3 Calculate the final pH of a solution made by mixing 25.0 mL of $0.0750 \text{ mol L}^{-1} \text{HCl}_{(\text{aq})}$ with 32.5 mL of $0.0675 \text{ mol L}^{-1} \text{KOH}_{(\text{aq})}$.



$$n(\text{HCl}) = c.V = 0.0750 \times 0.0250 = 1.875 \times 10^{-3} \text{ mol}$$

$$n(\text{KOH}) = c.V = 0.0675 \times 0.0325 = 2.194 \times 10^{-3} \text{ mol}$$

Enough information is known to calculate the moles ($n = c.V$) of HCl and KOH. This is a limiting reagent problem.

As the mole ratio from the balanced equation is 1:1, HCl is limiting reagent as $n(\text{HCl}) < n(\text{KOH})$.

$\therefore$ In the final solution, all HCl has been consumed and the pH is affected by $n(\text{KOH})$ not consumed.

$$\begin{aligned} n(\text{KOH}) \text{ not consumed} &= 2.194 \times 10^{-3} - 1.875 \times 10^{-3} \\ &= 3.188 \times 10^{-4} \text{ mol} \end{aligned}$$

$$\begin{aligned} c(\text{KOH}) &= n/V = 3.188 \times 10^{-4} / (0.0250 + 0.0325) \\ &= 5.54 \times 10^{-3} \text{ mol L}^{-1} \end{aligned}$$

$$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{5.54 \times 10^{-3}} = 1.80 \times 10^{-12} \text{ mol L}^{-1}$$

$$\text{pH} = -\log [\text{H}^+] = 11.7$$

5.2 THE pH SCALE

The pH of a solution is simply a measure of its acidity. It depends on the hydrogen ion concentration of the solution and is defined as follows.

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$\text{pH} < 7$	acidic
$\text{pH} = 7$	neutral
$\text{pH} > 7$	basic

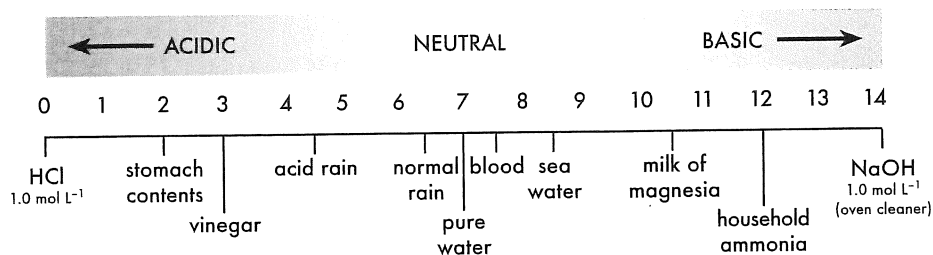


Figure 4.1 The pH acidity scale. As pH increases acidity decreases. An increase in pH by 1 means that a solution is only $\frac{1}{10}$ as acidic.

Question 5.1

Calculate the pH of 1.00 L of an antacid solution that contained 50.0 mg of $\text{Mg}(\text{OH})_2$ as its main ingredient.

Question 5.2

- (a) Determine the $[\text{H}^+]$ and the $[\text{OH}^-]$ of a sample of orange juice that had a pH of 3.60.

- (b) Determine the $[\text{H}^+]$ and $[\text{OH}^-]$ if the orange juice was then diluted by an equal volume of water (i.e. its volume was doubled using water).

Question 5.3

At 50.0°C, pure water was found to have a K_w of 5.48×10^{-14} . Determine the pH of the water at 50.0°C.

Question 5.4

A sample of sulfuric acid was found to have a pH of 2.50 at 50°C. Determine the $[\text{OH}^-]$ of this sample.

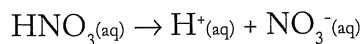
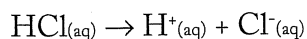
Question 5.5

Calculate the final pH of the solution formed when 47.5 mL of 0.540 mol L⁻¹ HNO₃ is mixed with 55.5 mL of 0.444 mol L⁻¹ NaOH.

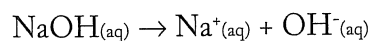
Defining Acids and Bases

Arrhenius Theory:

- acids provide hydrogen ions in aqueous solution

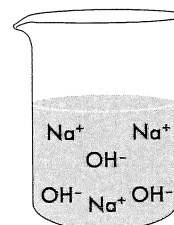
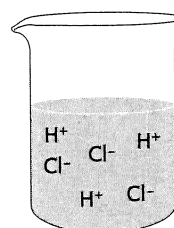
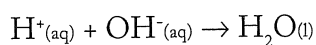


- bases provide hydroxide ions in aqueous solution



- neutralisation occurs when H^+ ion and OH^- ion combine to form $\text{H}_2\text{O}_{(\text{l})}$

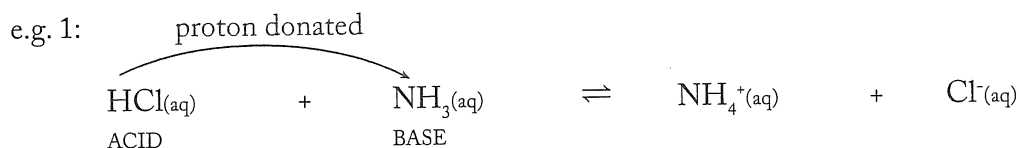
Neutralisation Reaction between $\text{HCl}_{(\text{aq})}$ and $\text{NaOH}_{(\text{aq})}$ is



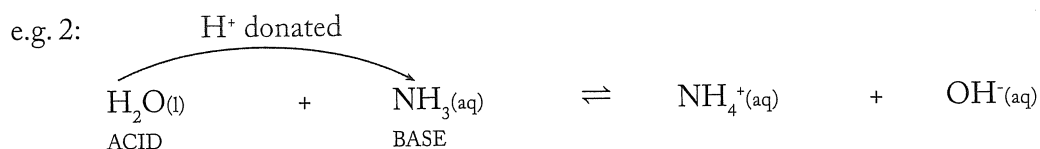
The Arrhenius model is restricted to aqueous solutions. A more general theory with wider application was proposed in 1923 by Johannes Brønsted (Denmark) and Thomas Lowry (England).

Brønsted-Lowry Theory

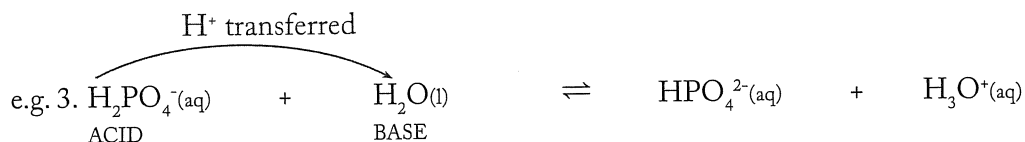
- acids act as proton (H^+) donors
- bases act as proton (H^+) acceptors



HCl is acting as an acid by donating a proton (H^+) to NH_3 . NH_3 , therefore, is acting as a base as it is accepting a proton.



H_2O is acting as an acid (proton donor)
 NH_3 is acting as a base (proton acceptor)



H_2PO_4^- is acting as an acid (donating a proton). H_2O is acting as a base (accepting a proton).

Note how H_2O can act either as an acid or a base using this theory (amphiprotic).

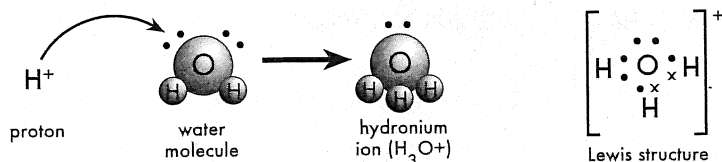
Question 5.6

Complete the following equation so as to show HCO_3^- acting as (a) an acid, (b) a base.



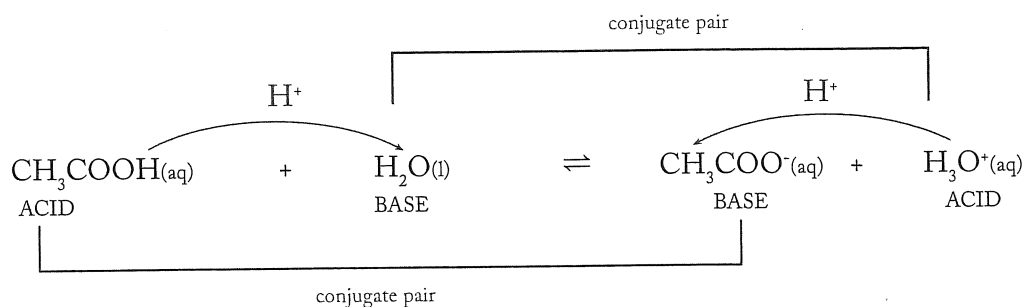
H^+ - A CLOSER LOOK!

The simplest atom, hydrogen, consists of a proton surrounded by an electron. Hence the symbol H^+ can be used to represent a **proton**. Protons are very small and readily attach themselves to the negative end of say, polar substances. In water they combine with H_2O to form the **hydronium ion**, H_3O^+ .



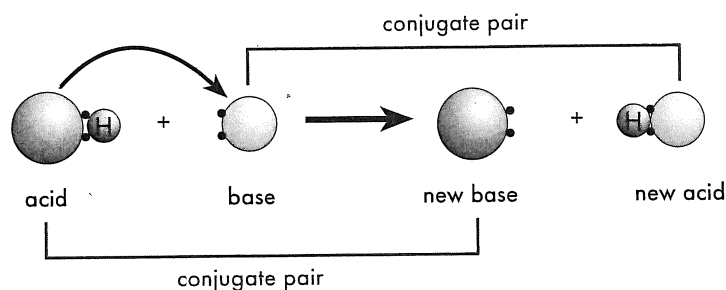
Conjugate acids and bases

When an acid loses (donates) a proton, what is left behind is its conjugate base. Similarly when a base accepts a proton, the substance formed is its conjugate acid.



i.e. $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$ and $\text{H}_3\text{O}^+/\text{H}_2\text{O}$

In general, the formation of conjugate pairs can be illustrated as follows.



Question 5.7

Complete the following tables. The first example in each case has been done for you.

(a)

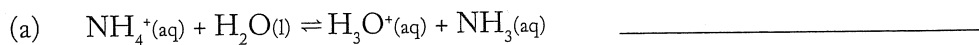
acid	conjugate base
HCl	Cl^-
H_2SO_4	
HF	
H_2S	
HCO_3^-	

(b)

base	conjugate acid
CO_3^{2-}	HCO_3^-
NH_3	
H_2PO_4^-	
F^-	
HSO_4^-	

Question 5.8

Identify the conjugate acid/base pairs in each of the following.

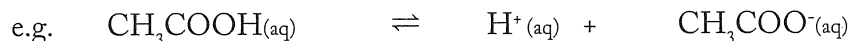


Strength of acids and bases

Strong acids completely ionise in water. For example, hydrogen chloride molecules in solution exist essentially as hydrogen ions, H^+ and chloride ions Cl^- .



Weak acids only partly ionise in water. For example, an ethanoic acid solution contains mostly CH_3COOH molecules.



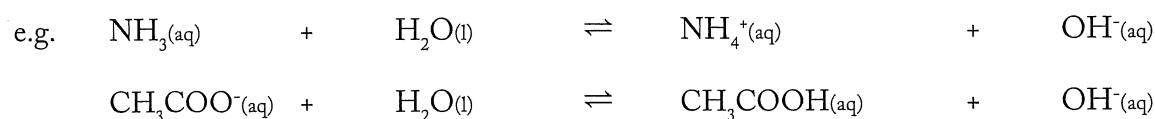
Notice the two arrows ($\rightleftharpoons$) are shown for the ionisation of weak acids.

Strong acids		Weak acids	
hydrochloric acid	HCl	oxalic acid	$\text{H}_2\text{C}_2\text{O}_4$
sulfuric acid	H_2SO_4	phosphoric acid	H_3PO_4
nitric acid	HNO_3	hydrofluoric acid	HF
hydrobromic acid	HBr	ethanoic acid	CH_3COOH
hydroiodic acid	HI	carbonic acid	H_2CO_3
perchloric acid	HClO_4	hypochlorous acid	HClO

Strong bases completely *dissociate* in water to produce hydroxide (OH^-) ions. Note that dissociation is simply the break up of ions that already exist within the ionic compound.



Weak bases only partially react with water in forming hydroxide (OH^-) ions. Apart from ammonia (NH_3), most bases are anions such as CO_3^{2-} and CH_3COO^- which react with water to produce OH^- ions.



Strong bases		Weak bases	
Soluble			
sodium hydroxide	NaOH	ammonia	NH_3
potassium hydroxide	KOH	carbonate ion	CO_3^{2-}
Slightly soluble		ethanoate ion	CH_3COO^-
calcium hydroxide	$\text{Ca}(\text{OH})_2$	fluoride ion	F^-
magnesium hydroxide	$\text{Mg}(\text{OH})_2$	phosphate ion	PO_4^{3-}

Hydrolysis of Salts

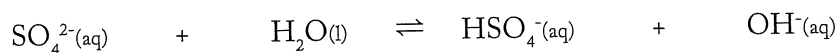
Hydrolysis is the process in which an *ion* reacts with water to produce H_3O^+ ions or OH^- ions. Ions that produce H_3O^+ are said to be acidic while ions that produce OH^- are said to be basic.

Acidic Anions:	HSO_4^- and H_2PO_4^- (i.e. anion from a polyprotic acid)
Basic Anions:	CH_3COO^- , HPO_4^{2-} , PO_4^{3-} , SO_4^{2-} , F^- , HCO_3^- , CO_3^{2-} , ClO^- , HS^- , CN^- (anion from a weak acid)
Neutral Anions:	Cl^- , NO_3^- , Br^- , I^- (anions from strong acids)
Acidic Cations:	NH_4^+ , Al^{3+} , Fe^{3+} (cation from a weak base or from aquated metal ions)
Basic Cations:	There are no basic cations covered in the syllabus
Neutral Cations:	Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Li^+ , Ba^{2+} (Group I and II metal cations)

Worked Example

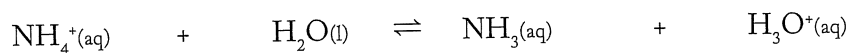
5.4 Write the equation for the hydrolysis of the following salt and classify the resulting solution as acidic or basic.

(a) Na_2SO_4 :



As OH^- ions are produced the solution is basic

(b) NH_4Cl :

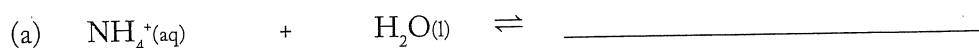


As H_3O^+ ions are produced, the solution is acidic

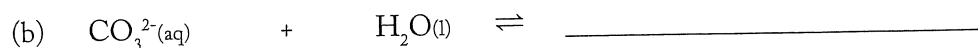
(NB: When writing an hydrolysis equation, the neutral anion or cation is a spectator ion and omitted)

Question 5.9

Complete the following hydrolysis reactions and indicate the type of solution they form (acidic or basic).



Solution becomes _____



Solution becomes _____



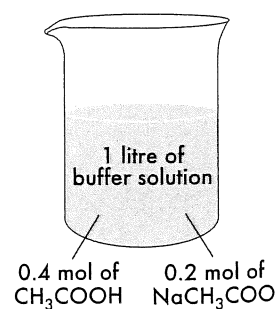
Complete the following table by predicting the nature of each salt solution as being acid, basic or neutral. The first is done for you.

Salt Solution	'Parent' base/acid	Nature
NH_4NO_3 (aq)	weak base/strong acid	acidic
NaCl (aq)		
Na_2CO_3 (aq)		
KH_2PO_4 (aq)		
MgF_2 (aq)		
$\text{Ca}(\text{ClO})_2$ (aq)		
K_2SO_4 (aq)		

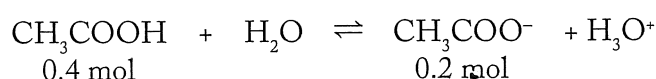
Buffer solutions are the mixture of a weak acid and its conjugate base or of a weak base and its conjugate acid. The advantage of buffer solutions is that they can maintain a relatively constant pH when quantities of acidic or basic materials are added to them. For a buffer to work, the amount of weak acid and its conjugate base must be large compared to the amount of H^+ and OH^- added or removed.

Buffer capacity is the ability of the buffer to resist changes in pH. It refers to the amount of H^+ or OH^- that can be added without causing a significant change to the pH of the buffer solution.

Buffering capacity increases as the concentration of the acid/conjugate base pair is increased. The greater the concentration of acid/conjugate base pair, the greater the amount of H^+ or OH^- that can be added without swamping the buffer's capacity to maintain a constant pH.



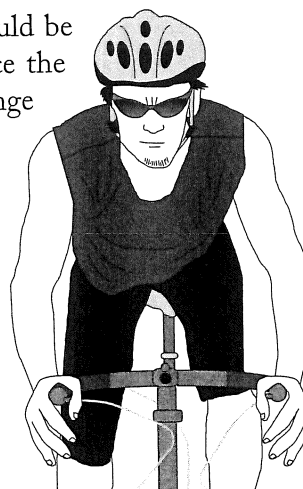
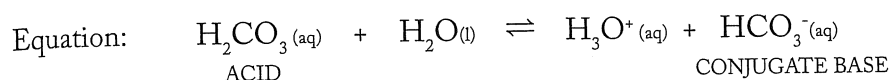
A buffer solution was made by dissolving enough CH_3COOH and NaCH_3COO in water to produce a solution that contained 0.4 mol of $\text{CH}_3\text{COOH}_{(\text{aq})}$ and 0.2 mol of $\text{CH}_3\text{COO}^{-}_{(\text{aq})}$.



The addition of only 0.01 mol of NaOH would neutralise 0.01 mol of H_3O^+ . The forward reaction would be favoured to partially counteract this change. As there is 0.4 mol of CH_3COOH available for the production of H_3O^+ (as per the forward reaction), the buffer would be operating within its capacity and be able to maintain a constant pH.

However if 0.5 mol of NaOH was added to the solution, the buffer would be swamped. It would not have sufficient moles of CH_3COOH to replace the H_3O^+ consumed by the NaOH. The pH of the buffer solution would change markedly.

Exercise tends to increase the acidity (reduce the pH) of blood. The presence of H_2CO_3 and its conjugate base, HCO_3^- help to maintain the bloods pH at 7.4.



During exercise more H_3O^+ is released into the blood from the muscles. Le Chatelier's Principle predicts that the reverse reaction would be favoured to partially counteract the increased $[\text{H}_3\text{O}^+]$ and effectively maintain a relatively constant blood pH.

Question 5.11

A second buffer solution in blood is the $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ combination.

- Write the equation for the reaction of the acidic ion, $\text{H}_2\text{PO}_4^-(\text{aq})$ with water.

- Identify the acid/base conjugate pairs in your equation.

- Use Le Chatelier's Principle to explain how this combination would respond to the addition of OH^- to the blood.

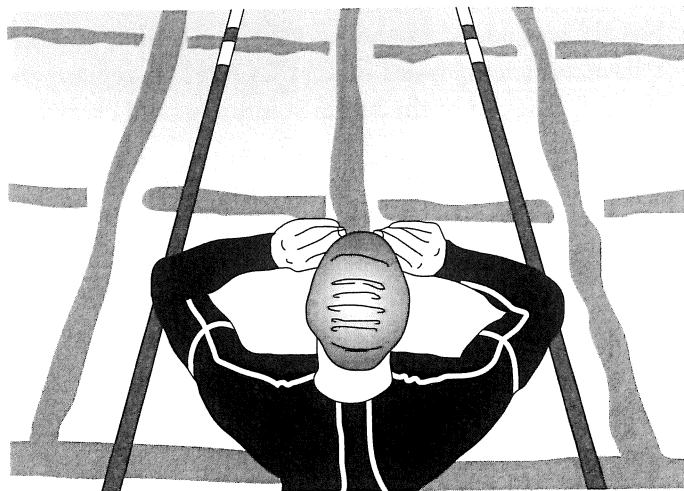
Question 5.12

Complete the following table by naming the conjugate base of the weak acids listed and then write the equation for the reaction of the weak acid with water.

WEAK ACID	CONJUGATE BASE	EQUATION
HF	F^-	$\text{HF}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$
CH_3COOH		
H_3PO_4		
NH_4^+		

Question 5.13

A 50,000 L swimming pool had a pH of 7.6.



- (a) What would be the expected new pH if 100 mL of 12.0 mol L^{-1} HCl was added to the pool?

- (b) After adding the 100 mL of 12.0 mol L^{-1} HCl to the pool, the owner measured the pH and found it to be 7.2. Account for the difference between the pH calculated in part (a) and the value obtained by the owner.

- (c) Suggest what chemicals could be added to the pool to form a buffer solution. Write the equilibrium equation for this buffering solution and suggest two points to consider when choosing a suitable buffer for a swimming pool.
